ECE 210 Midterm 3 Worksheet

Note: This worksheet is not guaranteed to be entirely representative of the midterm's contents. Material may appear on this worksheet which will not appear on the midterm, and vice versa.

Partial Fraction Expansion

1) (Technique 1)

a) sub problem 1

b) sub problem 2

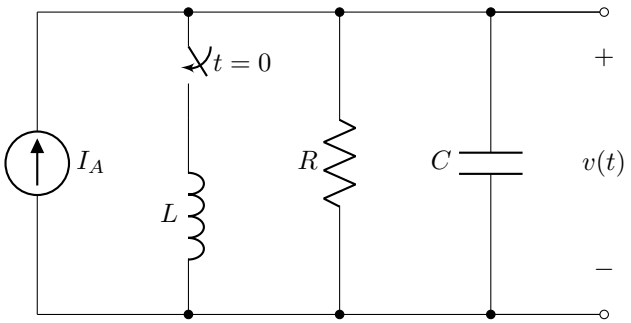
2) (Technique 2)

a) sub problem 1

b) sub problem 2

S-Domain Circuit Analysis

3) In the circuit below, the switch has been open for a long time and is closed at $t = 0$. Let $I_A = 1$ mA, $L = 2$ H, $R = 1.5$ kΩ, and $C = \frac{1}{6}$ μF.



a) Find the initial conditions $i_L(0^-)$ and $v(0^-)$, and draw the s-domain equivalent circuit for $t \geq 0$.

b) Find the zero-state and zero-input components of $V(s)$, and then find $v_{zs}(t)$ and $v_{zi}(t)$ for $t \geq 0$.

Marginal Stability & Resonance

4) Consider the causal linear time-invariant system

$$\frac{d^2y(t)}{dt^2} + \omega_0^2 y(t) = \omega_0^2 x(t), \qquad \omega_0 = 2 \text{ rad/s}.$$

The input and output are dimensionless, and t is measured in seconds. Use zero initial conditions in parts (b) and (c).

a) Find $H(s)$ and its poles. Find the zero-input response for arbitrary finite $y(0) = y_0$ and $y'(0) = v_0$. Is the natural response asymptotically stable, marginally stable, or unstable? Is the system bounded-input, bounded-output (BIBO) stable? Justify both answers.

b) Find the zero-state output for $x(t) = \cos(\Omega t)u(t)$, where $\Omega = 1$ rad/s. Is the output bounded? Does the natural-frequency component decay?

c) Repeat part (b) for $\Omega = \omega_0$. Explain resonance using the poles of $Y(s)$, and explain why the result is consistent with part (a).

5) problem 2

a) marginal stability part

b) resonance part

c) etc

Fourier Transforms

6) Find the Fourier transform:

$$g(t) = \frac{6}{t^2 + 9}$$

7) Find the output:

$$y(t) = e^{-2t}u(t) * \text{rect}(\frac{t}{2})$$